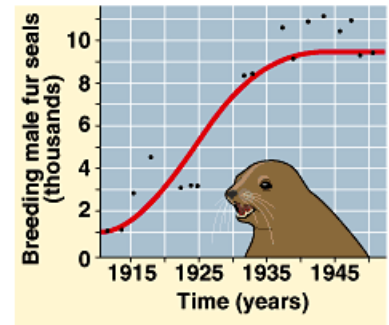


Take a deep breath. Relax. Choose the *best* answer. Clearly *circle* the letter for your answer. Vague or unclear marks will be counted as incorrect. Each question is worth 2 points.

1. After transient orca began to prey on sea otters, what happened to the kelp forest?
 - A. The kelp abundance decreased. Sea otters and kelp have a symbiotic relationship. Without the sea otters, the symbiotic relationship was broken and so the kelp abundance decreased.
 - B. The kelp abundance remained about the same. The kelp are not part of this community, so the kelp could not be affected.
 - C. The kelp abundance increased. Otters dislodge kelp with their playful antics. Because there were fewer otters, less kelp was dislodged so the kelp abundance increased.
 - D. The kelp abundance remained about the same. Only the predator-prey interactions between Orca and otters were important, so nothing happened to the kelp.
 - E. The kelp abundance decreased. Sea otters eat urchins and urchins eat kelp. Because there were fewer otters to eat urchins, the number of urchins increased. The urchins therefore ate more kelp so the kelp abundance decreased.

2. The type of growth curve shown in the figure below is a

- A. logistic growth curve.
- B. maximum growth curve.
- C. population density growth curve.
- D. carrying capacity growth curve.
- E. exponential growth curve.



3. On average, the amount of primary consumer biomass is about _____ less than the amount of primary producer biomass.
 - A. 10%
 - B. 50%
 - C. 20%
 - D. 90%
 - E. 80%
4. A transient killer whale eats seals. The seals ate fish. The fish ate part of a dead killer whale laying on the sea floor, but also ate smaller live squid. You do not know what the squid might have eaten. The fish in this example is acting as _____. (Choose the best answer)
 - A. a consumer and a decomposer.
 - B. a secondary consumer and a primary consumer.
 - C. a secondary consumer and a decomposer.
 - D. a secondary consumer, a decomposer, and a consumer one trophic level higher than the killer whale.
 - E. a consumer and a producer.

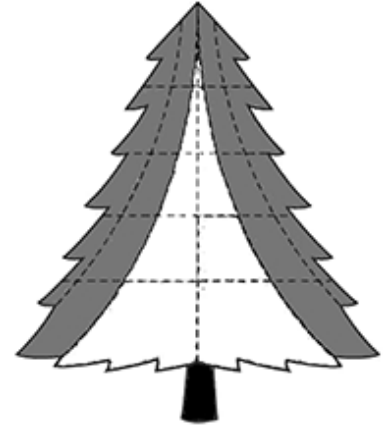
5. The land on the south side of the Himalayan mountain range (below) receives relatively high levels of rainfall and has fertile farmlands. The land north of the mountain range is hot and dry and can't be farmed. Which of the following is true.

- A. Warm, moist air from the Indian Ocean blows north, rises and cools, releasing precipitation as it moves up the south side of the range, and this cool, now dry air mass heats up as it descends on the north side of the range.
- B. The desert in southern China warms the north side of the mountain range, causing it to dry out.
- C. The south side of the Himalayan mountain range is in a rain shadow.
- D. The warm, moist air is too dense to rise over the height of Mount Everest and the other mountains in this range so the moisture remains on the south side of the range.
- E. The cool, moist Indian Ocean air heats up as it rises, releasing its precipitation as it passes the tops of the mountains, and this warm, now dry air cools as it descends on the leeward side of the range.



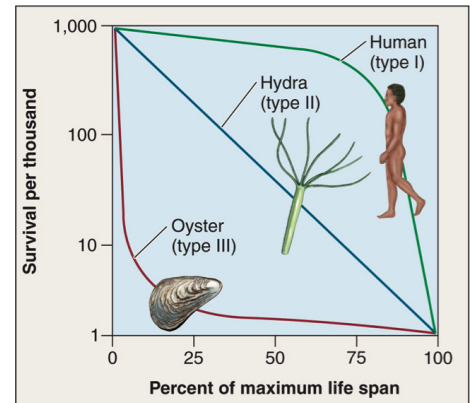
6. Considering your knowledge of the dead zone and the causes of microscopic algae blooms, the Gulf of Mexico ecosystem is emphmost likely to be limited by which of the following nutrients?
- A. hydrogen
 - B. nitrogen
 - C. carbon
 - D. CO₂
 - E. oxygen
7. Competition occurs between two different species in a community when
- A. both species are searching for mates.
 - B. both species need the same resources that is in limited supply.
 - C. one species is a predator of the other.
 - D. both species have partitioned resources.
 - E. one species is a parasite of the other.
8. The Mediterranean coast and coastal California have similar types of ecosystems. This is because
- A. they have similar types of organisms.
 - B. they are at similar latitudes.
 - C. they have similar climates.
 - D. they are both wine-growing regions.
 - E. they have similar types of communities.

9. When the number of individuals in a population remains more or less the same over time in the specific place where these individuals live, then this population has reached the _____ of that place.
- carrying capacity
 - logisitic growth capacity
 - maximum growth capacity
 - exponential growth capacity
 - population density
10. The “role” or “job” of a species in a community, represented by all of the resources needed by the species, is called
- the resource.
 - the niche.
 - a predator or prey.
 - a producer or consumer.
 - a predator.
11. Species Old can nest anywhere in a certain kind of pine tree. Recently, due to global climate change, Species New has arrived from down south. Species New also nests in the same kind of pine tree. However, Species New can only nest in the outermost parts of any branch (shown in darker color). Species Old chases away Species New in the inner part of the tree but ignores Species New in the outer part of the tree. What will *most likely* happen in the tree after a few years?
- Both species will nest in the outer part of the tree in equal numbers but only Species Old will nest in the inner part of the tree.
 - Species Old will prevent Species New from making nests so Species New will die out in this location.
 - Neither species will survive because they can’t nest in the entire tree.
 - Species New will prevent Species Old from making nests so Species Old will die out in this location.
 - Species Old will nest more often in the outer part of the tree.
 - Species New will nest in the outer part of the tree and Species Old will nest in the inner part of the tree.
12. Which of the following are important biotic (living) factors that can affect the organization of communities?
- light intensity, seasonality
 - nutrient availability, soil pH
 - predation, competition
 - precipitation, wind
 - temperature, water



13. According to the figure below, which is most correct about the offspring produced by Type I organisms?

- A. About the same number of offspring will die every year.
- B. Most offspring will die at a very young age.
- C. Most offspring will die about half way through their natural life.
- D. Most offspring will live most of their natural lifespan.
- E. Most offspring will die late in life.



14. Assuming a population has unlimited resources, exponential growth occurs because

- A. the immigration rate equals the death rate and the emigration rate equals the birth rate.
- B. the emigration rate exceeds the death rate.
- C. the birth rate exceeds the death rate.
- D. the emigration exceeds the birth rate.
- E. the death rate exceeds the birth rate.

15. In summer 2012, the Midwest experienced one of the worst droughts in recorded history. Which of the following statements is the best prediction of how this drought will have affected the Gulf of Mexico dead zone during summer 2012?

- A. The dead zone will be larger because the reduced volume of water will concentrate the nutrients that enter the Gulf of Mexico.
- B. The dead zone will be smaller because the Mississippi River will carry less water so fewer nutrients will be carried to the Gulf of Mexico.
- C. The dead zone will be smaller because less freshwater will enter the Gulf of Mexico, allowing more saltwater fishes to live around the mouth of the river.
- D. The dead zone will be similar to previous years because the drought is not likely to affect the amount of water that flows out through the Mississippi River.
- E. The dead zone will be smaller because farmers, anticipating the drought, would not have added fertilizer to the crop fields the spring before the drought.

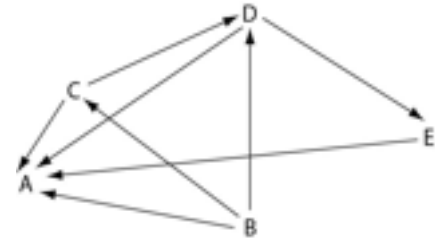
16. The dead zone in the Gulf of Mexico and other oceans around the globe occurs when

- A. an excess of nutrients causes an algal bloom. The algae consume all of the oxygen in the water for photosynthesis, so other organisms can no longer live there.
- B. an excess of nutrients causes an algal bloom, followed by death of the algae. Decomposition of the algae uses up the oxygen in the water, so other organisms can no longer live there.
- C. commercial fishers overharvest shrimp, crab, fishes and other sea animals.
- D. eutrophication causes an absence of nutrients in the zone.
- E. global climate change causes the water to become too warm to sustain life.

17. The difference between emigration and immigration is
- A. emigration is when individuals move out of a population; immigration is when individuals move out of a community.
 - B. the direction in which individuals are moving into or out of the population.
 - C. emigration is when individuals move into a population; immigration is when individuals move out of a population.
 - D. the overall movement of individuals among populations.
 - E. emigration is when individuals move out of a population; immigration is when individuals move into a population.
18. Consider the food chain grass → grasshopper → mouse → snake → hawk. How much of the chemical energy fixed by photosynthesis of the grass (100%) is available to the hawk?
- A. 0.1%
 - B. 10%
 - C. 1%
 - D. 0.01%
 - E. 60%
19. Carrying capacity
- A. is seldom reached by most population because the environment has plenty of resources.
 - B. is the maximum population size that a particular environment can support.
 - C. describes the stress a population undergoes due to limited resources.
 - D. does not for most species most of the time.
 - E. is determined by birth and death rates, and emigration and immigration rates.
20. Which of the following statements about symbiotic relationships is true?
- A. Neither organism benefits in a symbiotic relationship.
 - B. Both symbiotic organisms usually benefit from the relationship.
 - C. The most efficient type of parasite is one that kills its host.
 - D. Both organisms are harmed in a parasitic relationship.
 - E. A relationship that appears to be commensalism may in fact be mutualism or parasitism.
21. Bobcats eat forest mice. The mice eat seeds. Suppose a fire on the forest floor reduces the amount of seeds by 75%. Before the fire, the population of mice sustained a bobcat population of 8. What will *most likely* happen to the bobcat population as a result of the fire? The bobcat population will
- A. decrease by two but most will survive.
 - B. not be affected because the mice survived the fire.
 - C. increase as the bobcats shift to another food source.
 - D. decrease to two after a delay.
 - E. decrease to two almost immediately.
-

22. Based on this food web, which letter represents an organism that is a primary producer? The arrows point in the direction of energy flow (to higher trophic levels).

A. C
B. B
C. D
D. E
E. A



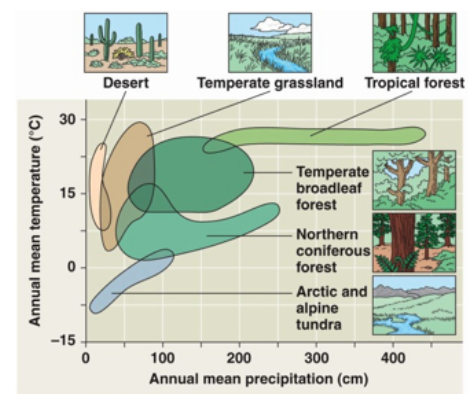
23. If humans stopped fishing along the coast of Alaska, what will probably happen to the kelp forest and why?
- A. The kelp abundance will increase because of the symbiotic relationship between kelp and fishes. Fishes are necessary for kelp to reproduce.
- B. The kelp abundance will decrease. Sea otters eat urchins and urchins eat kelp. Because there were fewer otters to eat urchins, the number of urchins increased. The urchins therefore ate more kelp so the kelp abundance decreased.
- C. The kelp abundance will decrease because more fish will be present to eat the kelp.
- D. The kelp abundance will increase because the sea urchin abundance will probably decrease.
- E. The kelp abundance will remain about the same. The kelp are not part of this community, so the kelp would not be affected.
24. A 20-year study of gray squirrels observed that the population size decreased during the first 15 years but increased during the last five years. With only this information, which of the following is the best explanation for the observed pattern?
- A. A better competitor, such as fox squirrels, prevented the gray squirrels from getting all of the food they needed to survive.
- B. The population was part of a trophic cascade.
- C. The effects of a severe drought caused a decrease in the amount of their normal food for 15 years but the squirrels eventually switched to a new food.
- D. The population size at the start of the study was greater than the carrying capacity but the population size eventually fell below carrying capacity.
- E. The population size of squirrel predators increased during the first 15 years and decreased during the last five years.
25. Some fish, like the goby in the picture below, live inside sponges. Sponges are a type of living organism. The goby benefits from the shelter provided by the sponge. The sponge receives no apparent benefit. As described here, this is an example of

A. symbiosis.
B. mutualism.
C. commensalism.
D. parasitism.
E. predation.



26. If immigration is greater than emigration and if the birth rate is greater than the death rate then
- the death rate will likely decrease and the birth rate will likely increase.
 - the population has plenty of resources.
 - the number of predators will also increase.
 - the population will experience exponential growth.
 - if nothing changes then the population will eventually reach or exceed the carrying capacity of the environment.
27. Plants produce water vapor during photosynthesis, which goes from the plant to the atmosphere. The water vapor in the atmosphere eventually falls back to the earth as rain. This process would be part of
- the energy cycle.
 - the water cycle.
 - the entire ecosystem.
 - plant metabolism.
 - energy flow.
28. Higher elevations on a mountain experience less precipitation and cooler temperatures than low elevations. As you descended from the top of the mountain, you would notice a change in the types of plants and animals present in the community. This is analogous to the changes in the ecosystems if you walked
- from the north pole to the south pole.
 - west to east at one latitude.
 - from the equator to the south pole.
 - east to west over different altitudes.
 - from the north pole to the equator.
29. The southeastern region of Missouri lies within the temperate broadleaf forest ecosystem. According to the climograph below, the temperate broadleaf forest

- is never warmer than the temperate grassland.
- has more rain and similar temperatures to the northern coniferous forest.
- always has more precipitation than the temperate grassland.
- has an average of 60–220 cm of precipitation per year.
- never has more than about 220 cm (87 inches) of rainfall per year.



30. Which of the following is *not* one of the four factors that play a role in population growth rate?
- A. Birth rate.
 - B. Demography.
 - C. Emigration.
 - D. Immigration.
 - E. Death rate.
-